

Study of a 30 MW bubbling fluidized bed combustor based on co-firing biomass and coal

The [original paper](#) contains 18 sections, with 9 passages identified by our machine learning algorithms as central to this paper.

Paper Summary

SUMMARY PASSAGE 1

Introduction

In the Asia-Pacific region, these residues are in plenty since a wide range of crops are grown to produce food for 56% of the world's population living in this part of the world [Natrajan et al., 1998]. Power generation based on biomass holds a considerable promise in Punjab, which has huge biomass resources from crop production system and agro industries. The total biomass production in Punjab was estimated at about 54.54 million tones [Kaur et al., 2006].

SUMMARY PASSAGE 2

Literature Review (Combustion Of Biomass And Coal):

A bench scale bubbling fluidized bed combustor consisting of a quartz glass tube of diameter 25 mm and height 650 mm has been used. The proportion of biomass with coal vary from 5% -25%. It describes five major issues associated with cofiring: ash deposition, SO₂ emission, NO_x formation, corrosion and carbon burnout.

SUMMARY PASSAGE 3

Experience At Cpp

In 2005 the plant worked on 100% usage of Rice Husk, for which they received successful CER verification. Rice husk is a processed fuel from industry and user friendly being clean and available in ready to use size (length= 6mm and width=0.5 to 1mm approximately) where as other biomass fuels are available in random sizes and require further processing to bring it to required size of 6mm (used at CPP). Due to this reason the number of buyers of rice husk has surpassed the number of suppliers in the region.

SUMMARY PASSAGE 4

Different Fuel Usage At Cpp

A few of biomass fuels used at CPP are shown in Fig. 2. Different biomass fuels are fed to the boiler with fuel feeders. While feeding biomass of different densities their calorific value has to be taken in to consideration.

SUMMARY PASSAGE 5

Challenges In Usage Of Biomass Fuels:

All the boilers are designed for biomass combustion. However in case of less availability of biomass fuel they may run on coal. Here a comparison of 100% rice husk usage (using data of 2005) has been done with cofiring and coal usage in terms of power generation and emissions.

SUMMARY PASSAGE 6

Challenges In Usage Of Biomass Fuels:

Test runs are not done for rice husk because it is not presently in use. During study of different features of biomass based boiler, it was observed that air consumption in case of biomass fuel combustion was more as compared to conventional fuel. The reason is that biomass fuel is having less (average) calorific value as compare to coal (Table 2), due to which the biomass is burnt in the freeboard region over the bed.

SUMMARY PASSAGE 7

Technology Evolved In Cpp:

CPP, Ambuja has implemented a number of these which are summarized as: i. Fuel conveying system was modified to cater to the needs of variety of biomass fuels as shown in Fig.6. ii. The boilers were modified to over bed fuel feeding mode to make them versatile for burning different types of biomass fuels. iii. On line soot blowers were installed to mitigate the fouling effects of ash.

SUMMARY PASSAGE 8

Analysis:

Radiation modeling and Non premixed combustion model for gaseous phase volatile combustion is the future scope of this work. Fig. 9 (a) is showing temperature contour of boiler working on cofiring, in which biomass is supplied from biomass ports at a height of 2m from the bed. Fig. 9(b) is showing temperature profile of another boiler running on coal.

SUMMARY PASSAGE 9

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The performance of the plant vis-À-vis efficiency and emissions should be described. The problems of using processed biomass of varying density that will influence the fluid bed operations (a feature that has received inadequate attention in the paper) should be brought out. It is entirely possible that at 30 % co-firing, the deviations are not substantive.